

ANALOG GUITAR EFFECTS CONSOLE



Cybertron

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- A fully analog guitar effects console that processes signals from a guitar or audio source through modules.



- Each effect can be individually bypassed
- The system delivers a high-quality analog output for amplification or recording.

Outcome of each sub-circuit

Pre-amplifier: Increase the input voltage to 1 V-2 V

Equalizer: Boost or cut specific frequency bands

Bass: 85Hz - 250 Hz

Mid: 250Hz - 1000Hz

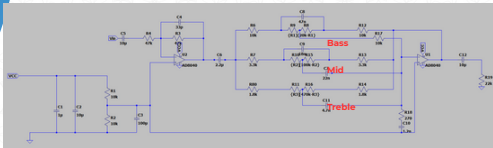
Treble: 1 kHz to 20 kHz

Tremolo: Generates Low frequency control signal and changes the signal amplitude accordingly

Distortion: Amplifies and clips the signal at 4 V

Tremolo Circuit

Equalizer Circuit



Equalizer Circuit

Bass: (Resistance from 0 to 20k - linearly varied)

3.45058238506
2.94360125065
2.52275240421
2.16973340511
1.86904877424
1.6079750061
1.38033860922
1.18064850569
1.00547903776
0.850183784962
0.712432652712

100 Hz

2.08457928896
1.83637356758
1.63145083189
1.46130150557
1.31373012066
1.18975687027
1.08049321175
0.984675467014
0.900928378105
0.830699443817
0.778349995613

215 Hz

1.35606133938
1.2171574831
1.10875204206
1.02398249507
0.957232445478
0.904615879059
0.864086031914
0.833368420601
0.81070381403
0.796066850424
0.802565932274

400 Hz

Equalizer Circuit

Mid: (Resistance from 0 to 100k - linearly varied)

2.06446313858
1.93043392897
1.73371779919
1.5497776866
1.39232438803
1.25479066372
1.131755054
1.01475259662
0.90420666337
0.800554037094
0.727868705988

200 Hz

4.63471198082
3.19697952271
2.47358870506
2.06271922588
1.78894388676
1.58137303591
1.40066945553
1.22408783436
1.03062272072
0.800157159567
0.551754504442

502 Hz

4.9167907238
3.04356324673
2.34176981449
1.97871845961
1.74859631062
1.57540041208
1.42414009571
1.26937359571
1.08698546886
0.842697292566
0.529841274023

796 Hz

3.89049077034
2.40440201759
1.91148090363
1.66953271627
1.52382051945
1.4192147851
1.32845878601
1.23849356174
1.12280040979
0.944391310215
0.640771329403

1261 Hz

Equalizer Circuit

Treble: (Resistance from 0 to 470k - linearly varied)

1.2872800231
1.21499049664
1.10632634163
1.04520452023
1.00620537996
0.977222532034
0.951944291592
0.925755888224
0.89302572608
0.846599519253
0.832494646311

700 Hz

1.63248288631
1.39783865213
1.24805021286
1.17974805832
1.1387591362
1.10926151276
1.08342576027
1.05532377958
1.01648533344
0.952425509691
0.926437795162

1109 Hz

2.21307766438
1.71944248751
1.51672083139
1.43942007761
1.39637964964
1.36470085382
1.33575612307
1.30394744873
1.25554218624
1.15906077623
1.07596975565

1758 Hz

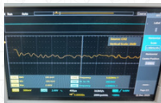
3.05595815182
2.02696204185
1.79688405991
1.71226793528
1.66377842426
1.62980318069
1.59596180916
1.5569756031
1.49660563469
1.36436313391
1.07898539305

2786 Hz

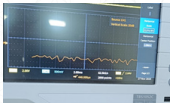
Equalizer Circuit



Bass

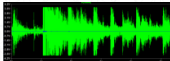
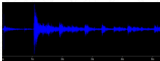
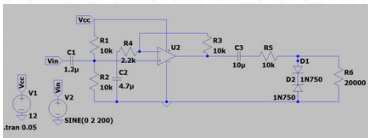


Mid



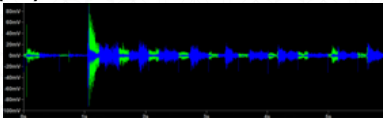
Treble

Distortion Circuit



Tremolo circuit

V(mV)



t(s)

Concept

- 555 timer in astable mode generates blinking LED.
- LED brightness modulates the LDR resistance.
- LDR acts as a variable resistor in the audio path produces the tremolo effect
- Two potentiometers are used for frequency controlling and Depth controlling separately.

Important Design Choices

- **555 Timer**

- Simple and inexpensive
- Easily configurable in astable mode
- Stable frequency
- Adjustable speed

- **Zener Diodes 3.3V**

- Low Zener Value for clipping
- The clipping effect can be controlled by increasing the gain.
- We use a soft clipping approach to pick up dynamics.

Important Design Choices

- **Op-Amp TL072CP and TL082CP**
 - Both are popular Low-Noise opamps
 - TL072CP is a single opamp, and TL082CP is a dual opamp
 - So both of them are used as required
 - Op-amp bandwidth: 80 Hz to 3 kHz , Input Amplitude: max 1.5 V
 - Expected Output Amplitude: 3-5 V
 - Gain-Bandwidth Product required: 0.064 MHz
 - SR required: 226.1 V/ms.

Resistance and Capacitance Values

- Equalizer - Based on Simulations
- Other Circuits: Based on Mathematical Equations

PCB Design (Done Using Altium Designer)



Enclosure Design

(Done Using SolidWorks)



Specifications

- Dimensions 160 mm X 100 mm X 53 mm
- Weight: 200 g
- Power Drawn at idle stage : 150mW
- Power Drawn by Power Amplifier: 5 mW

References

Equalizer:

<https://circuitdigest.com/electronic-circuits/audio-equalizer-tone-control-circuit-with-bass-treble-and-mid-frequency-control>

Pre-amp:

<https://circuitdigest.com/electronic-circuits/audio-equalizer-tone-control-circuit-with-bass-treble-and-mid-frequency-control>

Tremolo:

<https://curiousmotor.com/2020/06/04/simple-555-based-tremolo-effect/>



Thank You

